

# Introduction to Business Analytics and Descriptive Statistics

MGMT 670: Business Analytics



## Your part

- 3 HWs (work with teams)
- 1 Midterm (individual)
- 1 Final (individual)
- Textbook, software
- Reading/ discussion
- Details in syllabus

## Instructor

- Instructor: Will Wei Sun
  - Email: sun244@purdue.edu
  - Associate Professor of Management and Statistics (courtesy)
  - 2016-2019, Assistant Professor, Miami Business School
  - 2015-2016, Scientist, Advertising science team, Yahoo labs
  - 2011-2015, PhD in Stat & MS in Stat and CS, Purdue
- Class website: Brightspace
  - Slides, Reading materials
  - HW assignments
- The best way to reach me is via email!

## Advices

### 1. Keep up with materials

This is a **fast-paced** class. If unsure about anything ask questions in class and office hour.

### 2. Actively participate with your team

# Business Analytics

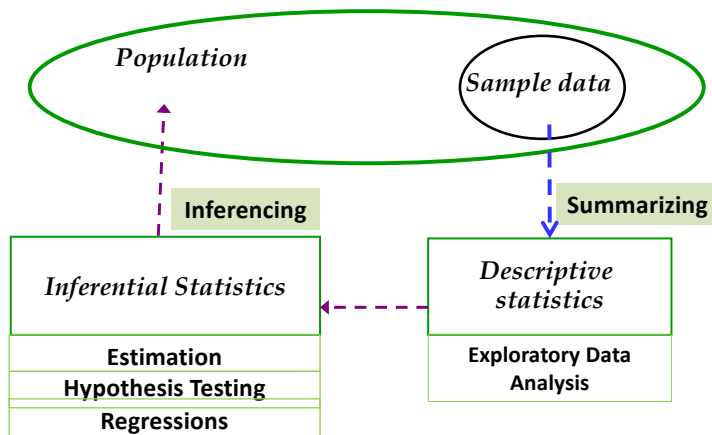
The science of collecting; organizing and presenting; analyzing and interpreting data for making business decisions.

# Applications:



- Finance: return and risk of portfolios; financial trends; predict future loan defaulters
- Marketing: consumer survey; advertising strategies; demand forecasting
- Production: quality control; product/process design; cost control
- Accounting: conducting audit sampling plan
- Human Resource: Recruiting, Salary, benefits

## Scheme for Statistical Analysis

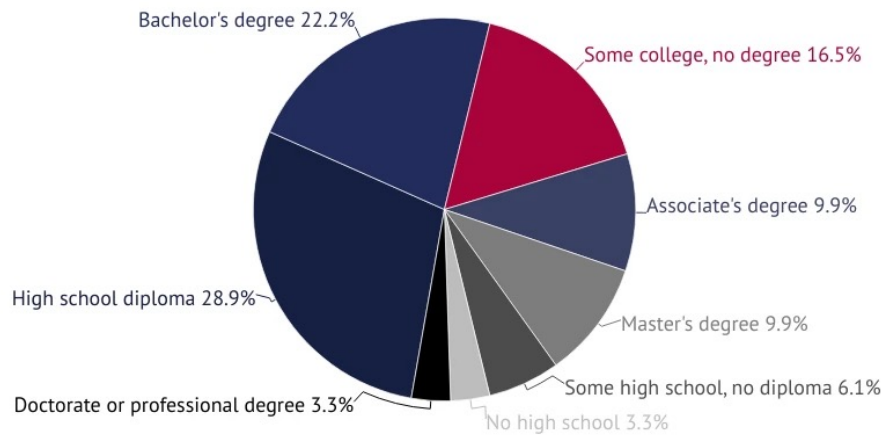


## Statistical Methods

- Descriptive Statistics
  - Tabular, graphical, and numerical methods used to summarize (or describe) data
  - Involves collecting, presenting, and characterizing data.
- Inferential Statistics
  - Process of using sample data to draw conclusions or make decisions about the characteristics of the population.

## Descriptive or Inferential Statistics?

### Highest Education Attainment Among Americans Aged 18+

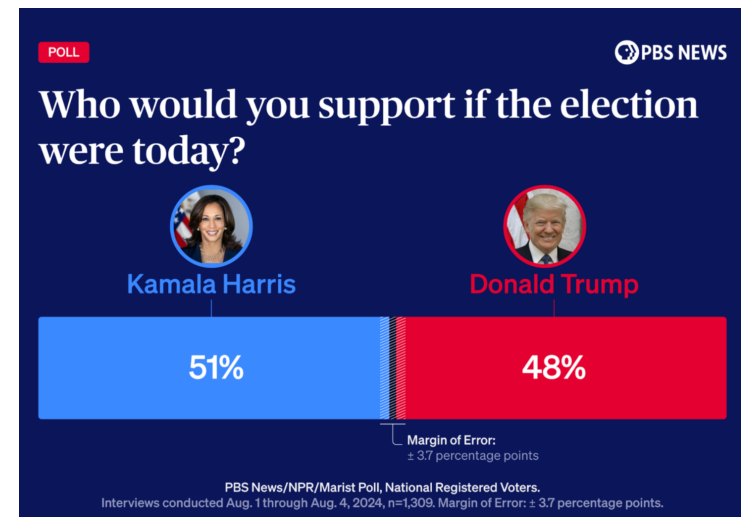


Education Data Initiative source: U.S. Department of Education National Center for Education Statistics

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## Descriptive or Inferential Statistics?



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## Topics in Descriptive Statistics

- Location measure
- Variability measure
- Shape of distribution
- Empirical Rule
- Association between Two Variables

### Minitab tools

- Summary stat.
- Box plot
- Scatter plot
- Correlation

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## Location measures

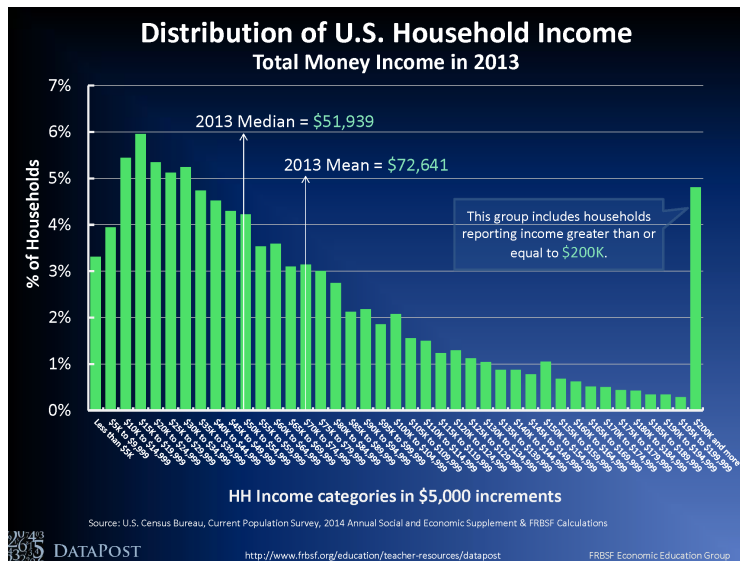
- Central Tendency: Mean and median  
What's the difference?

Which one to use when summarizing:

- Household income?
- Sales of a store?

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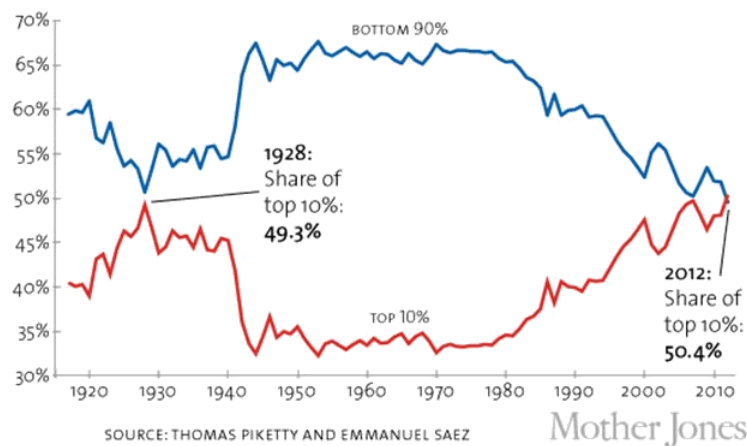
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<https://www.youtube.com/watch?v=QPKKQnijnsM>

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## Share of Total Income, 1917-2012 by percentile, including capital gains



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## Percentiles

- A data point that falls at the p-th percentile has a value greater than p% of the data points.
- Sort the data from smallest -> largest.
- Location of p-th percentile:  $L_p = \frac{p}{100}(n+1)$
- p-th percentile: the number at the location of  $L_p$

EXCEL (different): **p-th percentile** = percentile.exc(Data range, p%)

Median = percentile.exc(A1:A50, 0.5)

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## Example: Housing price

- A real estate agent has generated data on 80 houses that were recently sold in a college town. The selling prices (in thousands of dollars) are listed as follows.

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 101.93 | 104.01 | 105.06 | 105.07 | 106.11 | 109.25 | 111.56 | 114.09 | 114.34 | 116.34 |
| 116.44 | 117.53 | 117.87 | 118.09 | 118.13 | 118.17 | 118.30 | 119.37 | 119.65 | 119.70 |
| 120.15 | 121.27 | 121.99 | 122.14 | 122.57 | 123.48 | 123.49 | 124.56 | 126.30 | 126.72 |
| 127.15 | 127.20 | 127.35 | 127.59 | 127.97 | 128.50 | 128.91 | 129.15 | 129.23 | 129.91 |
| 130.31 | 130.39 | 130.44 | 130.98 | 131.33 | 131.47 | 132.05 | 132.24 | 132.42 | 133.17 |
| 133.22 | 133.94 | 134.10 | 134.21 | 134.33 | 134.61 | 135.70 | 136.01 | 136.57 | 136.89 |
| 137.08 | 137.44 | 137.54 | 137.66 | 137.74 | 138.56 | 139.53 | 140.04 | 140.13 | 140.16 |
| 140.93 | 141.10 | 141.13 | 141.63 | 141.78 | 141.81 | 141.90 | 142.87 | 143.12 | 185.58 |

## Example for determining quartiles

$Q_1 = 25^{\text{th}}$  percentile

$$L_{25} = (p/100)(n+1) = (25/100)(81) = 20.25$$

$$Q_1 = 119.70 + 0.25(120.15 - 119.70) = 119.8125$$

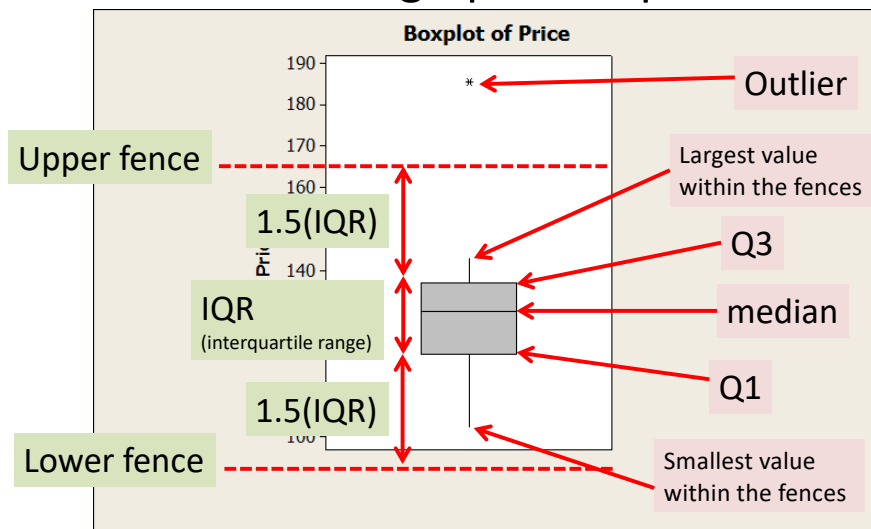
$Q_3 = 75^{\text{th}}$  percentile=?

$Q_2 = \text{median}=?$

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 101.93 | 104.01 | 105.06 | 105.07 | 106.11 | 109.25 | 111.56 | 114.09 | 114.34 | 116.34 |
| 116.44 | 117.53 | 117.87 | 118.09 | 118.13 | 118.17 | 118.30 | 119.37 | 119.65 | 119.70 |
| 120.15 | 121.27 | 121.99 | 122.14 | 122.57 | 123.48 | 123.49 | 124.56 | 126.30 | 126.72 |
| 127.15 | 127.20 | 127.35 | 127.59 | 127.97 | 128.50 | 128.91 | 129.15 | 129.23 | 129.91 |
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| 133.22 | 133.94 | 134.10 | 134.21 | 134.33 | 134.61 | 135.70 | 136.01 | 136.57 | 136.89 |
| 137.08 | 137.44 | 137.54 | 137.66 | 137.74 | 138.56 | 139.53 | 140.04 | 140.13 | 140.16 |
| 140.93 | 141.10 | 141.13 | 141.63 | 141.78 | 141.81 | 141.90 | 142.87 | 143.12 | 185.58 |

## Box-and-Whisker Plot

Minitab: graph->boxplot



## Variability measures:

- Inter-quartile range : IQR
- Variance
- Standard deviation

## Population variance & population standard deviation

$$\text{Variance: } \sigma^2 = \sum \frac{(x_i - \mu)^2}{N}$$

$$\text{Stdev: } \sigma = \sqrt{\sum \frac{(x_i - \mu)^2}{N}}$$

- Example: -100,-50,-20, 20, 50,100
- Variance= average squared distance from the mean=?
- Stdev= average distance from the mean=?

## We actually use different measures!

$$\text{sampled variance: } s^2 = \sum \frac{(x_i - \bar{x})^2}{n-1}$$

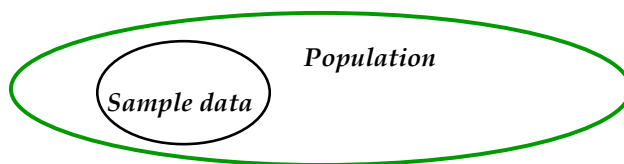
$$\text{sampled stdev: } s = \sqrt{\sum \frac{(x_i - \bar{x})^2}{n-1}}$$

WHY????



correct for the bias introduced by using the sample mean as an estimate of the population mean

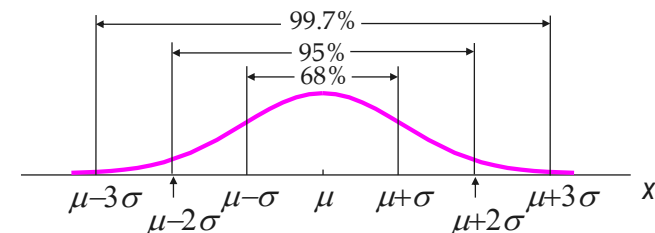
## Standard notations



| Measures           | Sample    | Population |
|--------------------|-----------|------------|
| Mean               | $\bar{x}$ | $\mu$      |
| Standard Deviation | $s$       | $\sigma$   |
| Variance           | $s^2$     | $\sigma^2$ |
| Size               | $n$       | $N$        |

## The Empirical Rule

- For data having a bell-shaped (normal) distribution:



If we take a random sample (number) from this distribution, each percentage is the chance that the sample will fall in the corresponding range.

## Example: Housing Price

For housing data, 77/80 = 96.25% of the prices are between 104.19 and 153.27.

$$\bar{x} = 128.73$$

$$s = 12.27$$

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 101.93 | 104.01 | 105.06 | 105.07 | 106.11 | 109.25 | 111.56 | 114.09 | 114.34 | 116.34 |
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| 133.22 | 133.94 | 134.10 | 134.21 | 134.33 | 134.61 | 135.70 | 136.01 | 136.57 | 136.89 |
| 137.08 | 137.44 | 137.54 | 137.66 | 137.74 | 138.56 | 139.53 | 140.04 | 140.13 | 140.16 |
| 140.93 | 141.10 | 141.13 | 141.63 | 141.78 | 141.81 | 141.90 | 142.87 | 143.12 | 185.58 |

## Relative Location from Mean: z-Scores

- A z-score is often called the standardized value:

$$z_i = \frac{x_i - \bar{x}}{s}$$

- Denotes the number of standard deviations a value  $x_i$  is from the mean.
- Signs: - (below mean) and + (above mean).

## Example: z- Score of Housing Price

|       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| -2.18 | -2.01 | -1.93 | -1.93 | -1.84 | -1.59 | -1.40 | -1.19 | -1.17 | -1.01 |
| -1.00 | -0.91 | -0.89 | -0.87 | -0.86 | -0.86 | -0.85 | -0.76 | -0.74 | -0.74 |
| -0.70 | -0.61 | -0.55 | -0.54 | -0.50 | -0.43 | -0.43 | -0.34 | -0.20 | -0.16 |
| -0.13 | -0.13 | -0.11 | -0.09 | -0.06 | -0.02 | 0.01  | 0.03  | 0.04  | 0.10  |
| 0.13  | 0.13  | 0.14  | 0.18  | 0.21  | 0.22  | 0.27  | 0.29  | 0.30  | 0.36  |
| 0.37  | 0.42  | 0.44  | 0.45  | 0.46  | 0.48  | 0.57  | 0.59  | 0.64  | 0.66  |
| 0.68  | 0.71  | 0.72  | 0.73  | 0.73  | 0.80  | 0.88  | 0.92  | 0.93  | 0.93  |
| 0.99  | 1.01  | 1.01  | 1.05  | 1.06  | 1.07  | 1.07  | 1.15  | 1.17  | 4.63  |

$$\bar{x} = 128.73$$

$$s = 12.27$$

## Distribution Rules

### ■ Chebyshev's Theorem

- At least  $(1 - 1/k^2)$  of the items in **any data set** are within  $k$  standard deviations of the mean,  $k > 1$ .

— [https://en.wikipedia.org/wiki/Chebyshev%27s\\_inequality](https://en.wikipedia.org/wiki/Chebyshev%27s_inequality)

### ■ Empirical Rule

- For data having a bell-shaped distribution

## Chebyshev's Theorem

At least  $(1 - 1/k^2)$  of the items in any data set will be within  $k$  standard deviations of the mean, where  $k > 1$ .

### Example:

- At least **75%** of the items must be within  $k = 2$  standard deviations of the mean.
- At least **89%** of the items must be within  $k = 3$  standard deviations of the mean.
- At least **94%** of the items must be within  $k = 4$  standard deviations of the mean.

## Example: Housing Price

### ■ Chebyshev's Theorem

Let  $k = 2.0$  with  $\bar{x} = 128.73$  and  $\sigma = 12.27$

At least  $(1 - 1/2^2) = 0.75$  or 75% of the rent values must be between

$$\bar{x} - k\sigma = 128.73 - 2.0(12.27) = 104.19$$

and

$$\bar{x} + k\sigma = 128.73 + 2.0(12.27) = 153.27$$

## Detecting Outliers

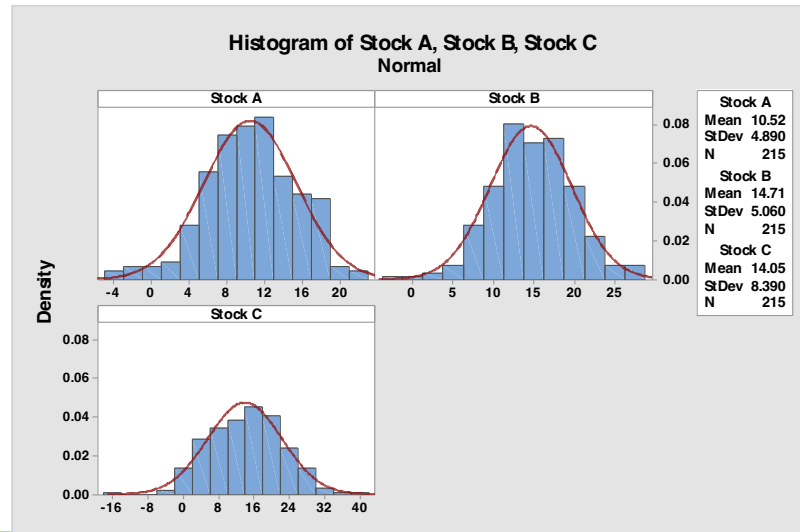
- It might be an incorrectly recorded/included data value. **Examples?**
- It might be a correctly recorded data value that belongs in the data set. **Examples?**
- An outlier is an unusually small or unusually large value in a data set.
  - A data value with a z-score less than -3 or greater than +3 might be considered an outlier.

## Summary of this lecture

- Mean vs median
- Percentiles and BoxPlot
- Population Stdev vs sampled Stdev
- Empirical rule
- Z-score

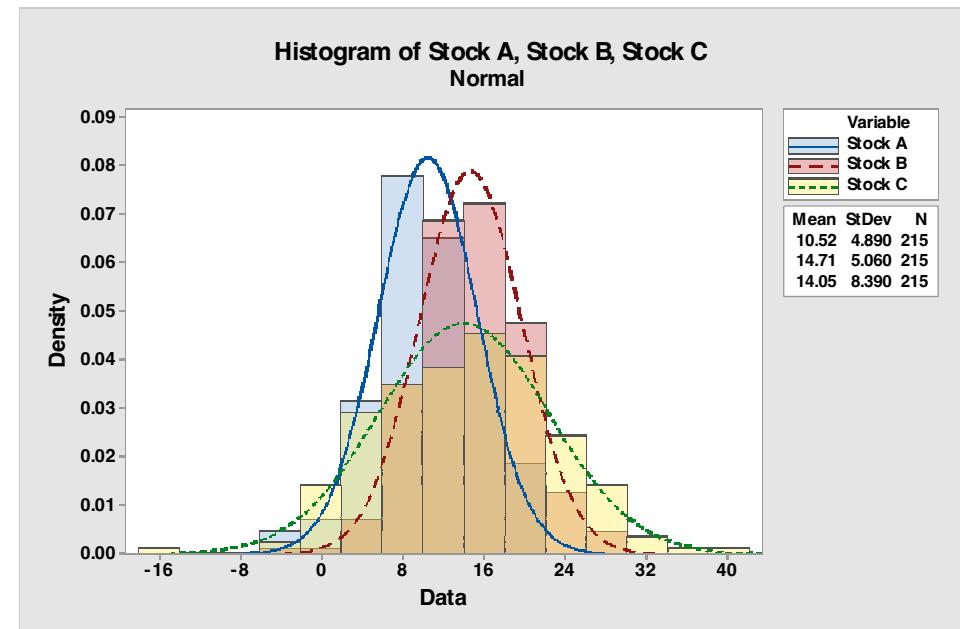


## Compare stocks based on the data of their return



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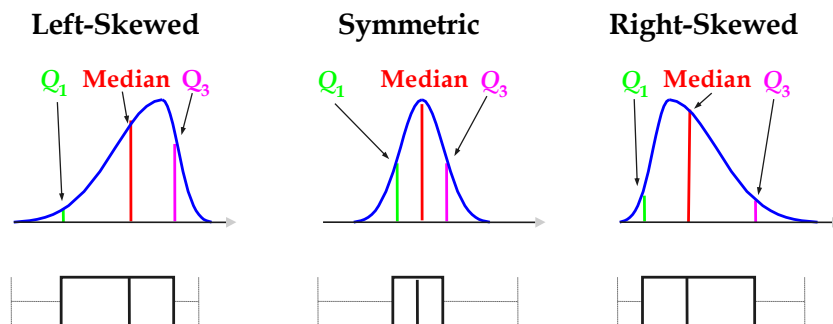
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## Correspondence between a Population Distribution and a Sample Box-plot



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## Association between Two Variables

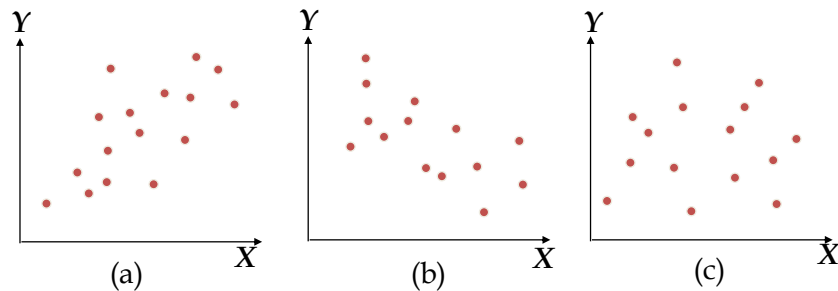
- Often a manager is interested in the relationship between two variables.
  - (Advertising expenditure, Sales)
  - (Input, Output)
  - (Income, Credit rating)
  - Data have to be collected in pair:
  $(X_i, Y_i), i = 1, \dots, n$
  - Tools:
    - Graphical/Tabular: Scatter plot, Contingency table
    - Numerical: Covariance, Correlation

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# Scatter Plot

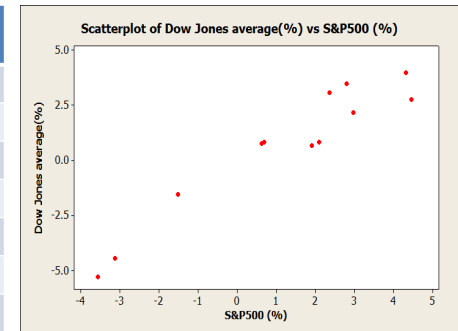
- Graphical representation of the relationship between two quantitative variables  $X$  and  $Y$
- Examples



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| month          | S&P500 (%) | Dow Jones average(%) |
|----------------|------------|----------------------|
| July 31, 2014  | -1.51      | -1.56                |
| June 30, 2014  | 1.91       | 0.65                 |
| May 31, 2014   | 2.10       | 0.82                 |
| April 30, 2014 | 0.62       | 0.75                 |
| Mar. 31, 2014  | 0.69       | 0.83                 |
| Feb. 28, 2014  | 4.31       | 3.97                 |
| Jan. 31, 2014  | -3.56      | -5.30                |
| Dec. 31, 2013  | 2.36       | 3.05                 |
| Nov. 30, 2013  | 2.80       | 3.48                 |
| Oct. 31, 2013  | 4.46       | 2.75                 |
| Sept. 30, 2013 | 2.97       | 2.16                 |
| Aug. 31, 2013  | -3.13      | -4.45                |



- Co-vary-> Covariance
- Positive relationship
- Example of negative relationship?
- Minitab -> Graph tab

EXAMPLES, PRACTICE QUESTIONS

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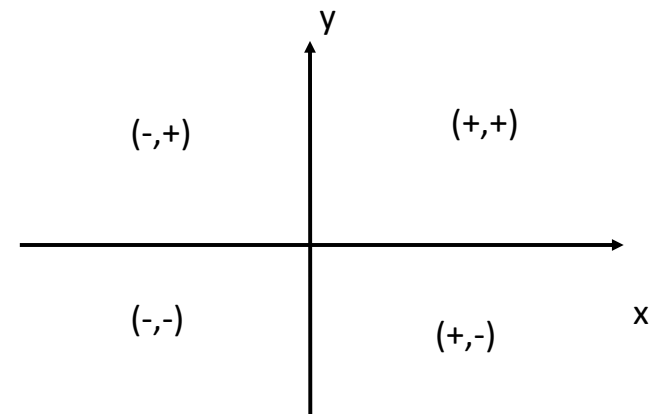
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# Covariance

- Measure of the **linear association** between two variables.
- Positive values indicate a positive relationship.
- Negative values indicate a negative relationship.

- DIRECTION +

**NOT about strength of the relationship!**



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EXAMPLES, PRACTICE QUESTIONS

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## Covariance

- Sample covariance:

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

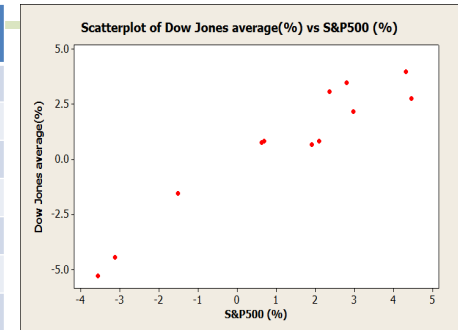
- Population covariance:

$$\sigma_{xy} = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

EXCEL: Sample covariance = **COVARIANCE.S** (X-range, Y-range)  
 Population covariance = **COVARIANCE.P** (X-range, Y-range)  
 Minitab: Stat -> Basic Statistics -> Covariance

## How to understand this formula?

| month          | S&P500 (%) | Dow Jones average(%) |
|----------------|------------|----------------------|
| July 31, 2014  | -1.51      | -1.56                |
| June 30, 2014  | 1.91       | 0.65                 |
| May 31, 2014   | 2.10       | 0.82                 |
| April 30, 2014 | 0.62       | 0.75                 |
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| Nov. 30, 2013  | 2.80       | 3.48                 |
| Oct. 31, 2013  | 4.46       | 2.75                 |
| Sept. 30, 2013 | 2.97       | 2.16                 |
| Aug. 31, 2013  | -3.13      | -4.45                |



How to calculate this in Excel?

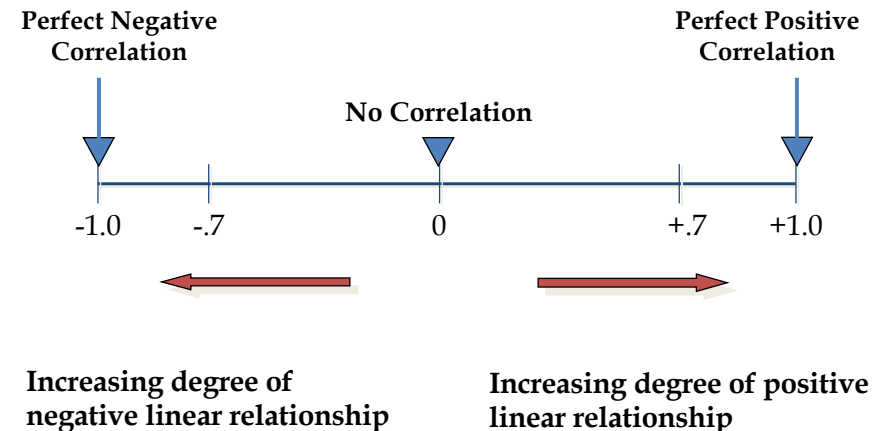
## Correlation Coefficient

- Unit-free measure of degree of linear association between two variables
- Correlation coefficient:

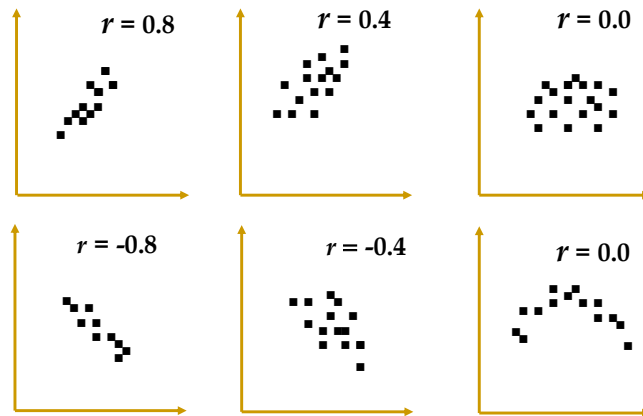
$$r_{xy} = \frac{s_{xy}}{s_x s_y}$$

EXCEL: Correlation coefficient = **CORREL** (X-range, Y-range)  
 Minitab: Stat -> Basic Statistics -> Correlation

## Values of Correlation Coefficients



## Correlations and Scatter Plots



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## Correlation Caveats

- Correlation is only applicable to **LINEAR relationship**. There are many other types of relationships
- Correlation is not Causation (Spurious Correlations)

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## Spurious Correlations

- <https://www.tylervigen.com/spurious-correlations>

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## Case Study: Auto parts



EXAMPLES, PRACTICE QUESTIONS

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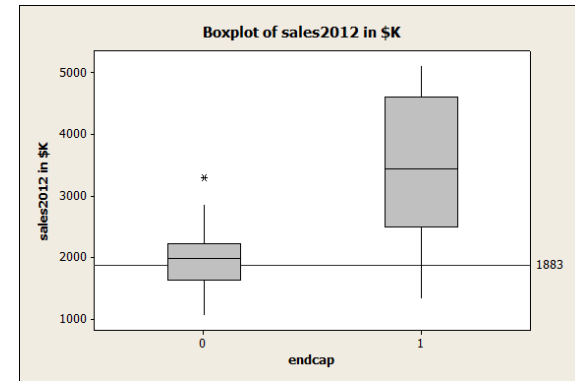
## Auto parts

- Auto-parts store signs up for an end-cap location in U-shaped mall
- New building blocks view from street
- After operating for a year, Auto-parts store sues real estate developer
- For this store:

|                                 |           |
|---------------------------------|-----------|
| 2012 Sales                      | 1,883,000 |
| Population in 3-mile radius     | 66,000    |
| Average income in 3-mile radius | 70,000    |
| Yelp ratings                    | 2 stars   |

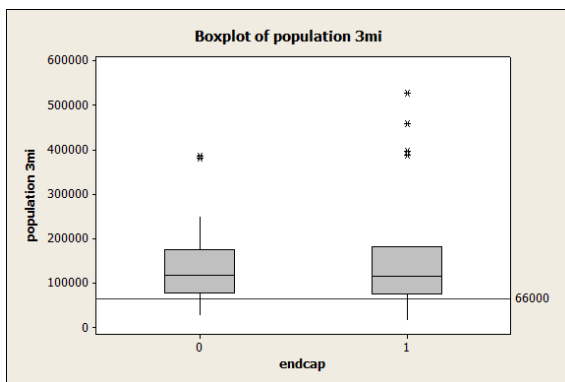
- Data for 45 stores is available from the same state, 23 of which are end-cap.

## Box Plot of Sales2012



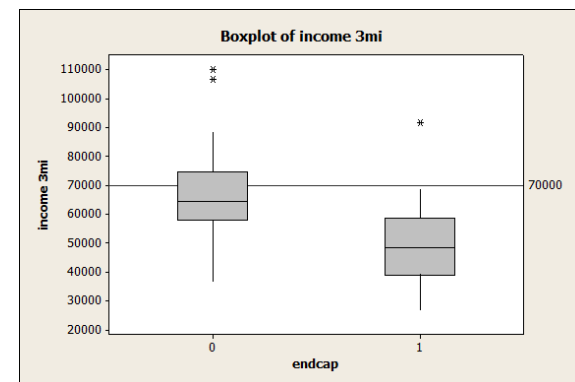
Minitab : Graph -> Boxplot -> One Y, Group -> .... ; To add lines: Scale-> Reference Lines

## Box Plot of Population



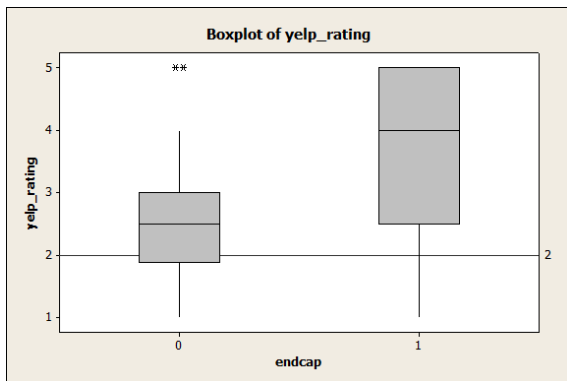
Minitab : Graph -> Boxplot -> One Y, Group -> ....

## Box Plot of Income



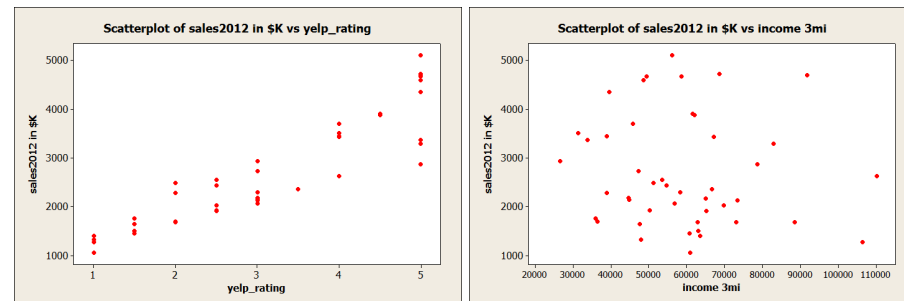
Minitab : Graph -> Boxplot -> One Y, Group -> ....

## Box Plot of Yelp Rating



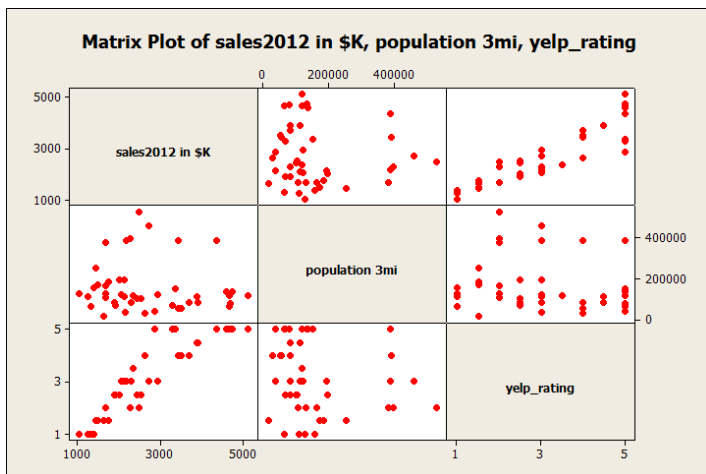
Minitab : Graph -> Boxplot -> One Y, Group -> ....

## Minitab: Scatter Plot

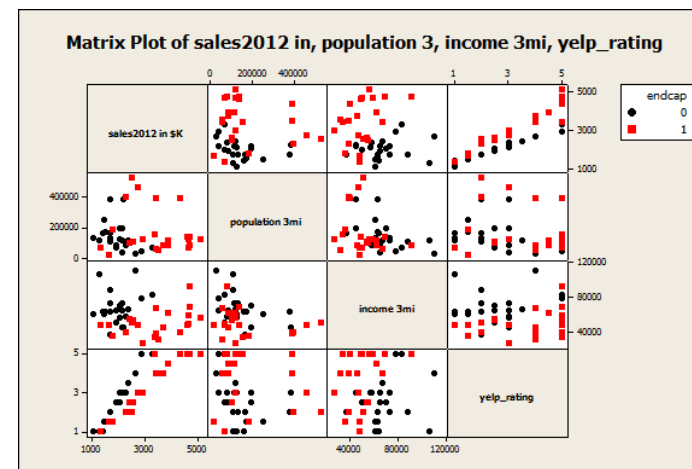


Minitab: Graphs -> Scatterplot -> Simple -> ....

## Matrix Plot



## Matrix Plot



Minitab: Graphs -> Matrix Plot -> With Groups -> ....

# Covariances and Correlations

Correlations: sales2012 in \$K, population 3mi, income 3mi, yelp\_rating

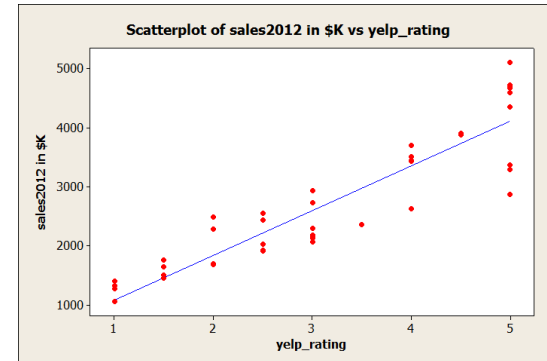
|                | sales2012 in \$K | population 3mi | income 3mi |
|----------------|------------------|----------------|------------|
| population 3mi | -0.071           |                |            |
| income 3mi     | -0.104           | -0.343         |            |
| yelp_rating    | 0.913            | -0.164         | 0.009      |

Covariances: sales2012 in \$K, population 3mi, income 3mi, yelp\_rating

|                  | sales2012 in \$K | population 3mi | income 3mi | yelp_rating |
|------------------|------------------|----------------|------------|-------------|
| sales2012 in \$K | 1269092          |                |            |             |
| population 3mi   | -9924663         | 15589672446    |            |             |
| income 3mi       | -2133137         | -783894622     | 334571350  |             |
| yelp_rating      | 1398             | -27862         | 224        | 2           |

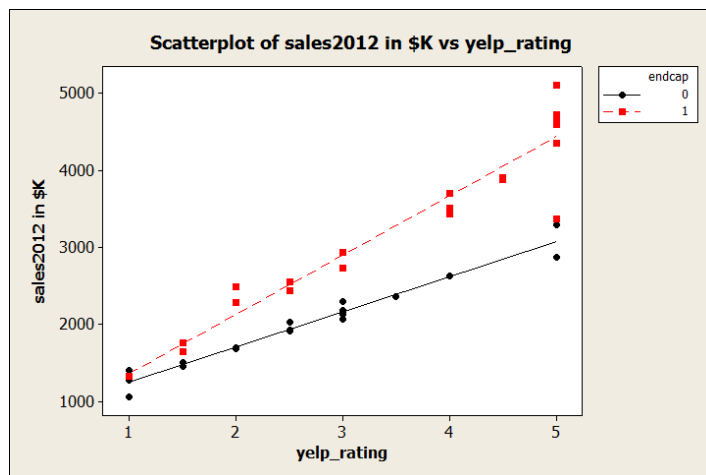
Minitab: Stat -> Basic Statistics -> Correlation or Covariance -> ...

# Scatter plot with regression



– Use regression line to predict;

# Scatter Plot (by location type)



Minitab: Graph -> Scatterplot -> With Regression and Groups -> ....

# Summary

- Location measure
- Variability measure
- Empirical Rule
- Association between Two Variables

## Minitab tools

- Summary stat.
- Box plot
- Scatter plot
- Correlation